Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech
Code: 102601

Subject: Design of Machine Elements

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*
(v) Standard design data book is allowed.

Q.1 Choose the correct option / answer the following (Any seven question only): [2 x 7 = 14]

(a) Stress concentration in a machine component of ductile material is not so harmful as it is in brittle material because
- (i) In ductile material local yielding may distribute stress concentration
- (ii) Ductile material has larger Young's modulus
- (iii) Poisson's ratio is larger in ductile material
- (iv) Modulus of rigidity is larger in ductile material

(b) If the size of a standard specimen for a fatigue testing machine is increased, the endurance limit for the material will
- (i) Have same value as that of standard specimen
- (ii) Decrease
- (iii) Increase
- (iv) None of the above

(c) Two solid shafts A and B are made of the same material, and A is twice the diameter of B. The torque that can be transmitted by A is
- (i) 2 times that of B
- (ii) 4 times that of B
- (iii) 6 times that of B
- (iv) 8 times that of B

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(d) If the load on a ball bearing is halved, its life
- (i) Remains unchanged
- (ii) Increases two times
- (iii) Increases four times
- (iv) Increases eight times

(e) Which of the following is not a type of welding joint?
- (i) Lap joint
- (ii) Butt joint
- (iii) Tee joint
- (iv) Single strap joint

(f) A brake commonly used in motor cars is
- (i) shoe brake
- (ii) band brake
- (iii) band and block brake
- (iv) internal expanding brake

(g) If Z = Absolute viscosity of the lubricant in kg/m-s. N = Speed of the journal in r.p.m., and p = Bearing pressure in N/mm^2, then the bearing characteristic number is
- (i) Z N/p
- (ii) Z p/N
- (iii) Z/p N
- (iv) p N/Z

(h) In leaf springs, the longest leaf is known as
- (i) lower leaf
- (ii) master leaf
- (iii) upper leaf
- (iv) none of these

(i) Lewis equation in spur gears is used to find the
- (i) tensile stress in bending
- (ii) shear stress
- (iii) compressive stress in bending
- (iv) fatigue stress

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(j) Which of the following is antifriction bearing?

(i) journal bearing
(ii) pedestal bearing
(iii) collar bearing
(iv) needle bearing

[7]

Q.2 (a) Explain the general considerations in machine design. [7]

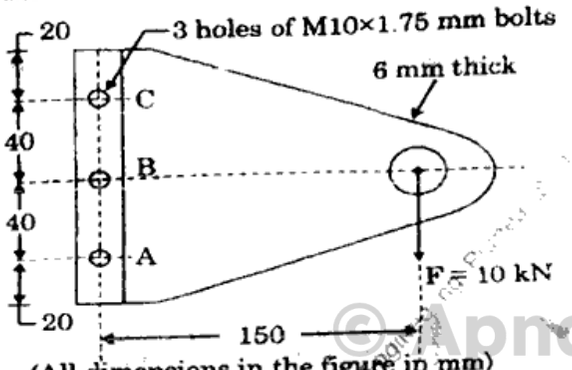
(b) A rotating steel shaft is supported at the ends. It is subjected to a point load at the center. The maximum bending stress developed is 100 MPa. If the yield, ultimate & corrected endurance strength of the shaft material is 300 MPa, 500 MPa & 200 MPa respectively, find the factor of safety for the shaft. [7]

Q.3 (a) Explain wedge film and squeeze film journal bearings. [7]

(b) A lightly loaded full Journal bearing has Journal diameter of 50 mm, bush bore of 50.05 mm and bush length of 20 mm. If rotational speed of Journal is 1200 rpm and average viscosity of liquid lubricant is 0.03 Pa s, calculate the power loss. [7]

Q.4 (a) Discuss the failure of a riveted joint under different stresses. [7]

(b) A bolted joint is shown in the figure below. Calculate the maximum shear stress in the bolts at A and B.



(All dimensions in the figure in mm)

Q.5 (a) With the help of neat sketches, differentiate between single and multi plate clutches. [7]

(b) A single plate clutch, effective on both sides, is required to transmit 25 kW at 3000 r.p.m. Determine the outer and inner diameters of frictional surface if the coefficient of friction is 0.255, ratio of diameters is 1.25 and the maximum pressure is not to exceed 0.1 N/mm^2. Also determine the axial thrust to be provided by springs. Assume the theory of uniform wear. [7]

Q.6 (a) A bearing supports a radial load of 7000 N and a thrust load of 2100 N. The desired life of the ball bearing is $160 \cdot 10^6$ revolutions at 300 rpm. If the load is uniform and steady, service factor is 1, radial factor is 0.65, thrust factor is 3.5, k = 3 and rotational factor is 1. Calculate the basic dynamic load rating of a bearing. [7]

(b) A 60 mm long and 6 mm thick fillet weld carries a steady load of 15 kN along the weld. The shear strength of the weld material is equal to 200 MPa. Calculate the factor of safety. [7]

Q.7 (a) Derive the Lewis Equation for the beam strength of a gear tooth. [7]

(b) A bronze spur pinion rotating at 600 r.p.m. drives a cast iron spur gear at a transmission ratio of 4: 1. The allowable static stresses for the bronze pinion and cast iron gear are 84 MPa and 105 MPa respectively. The pinion has 16 standard 20° full depth involute teeth of module 8 mm. The face width of both the gears is 90 mm. Find the power that can be transmitted from the standpoint of strength. Tooth form factor is given for 0° full depth involute $Y = 0.154 - \frac{0.912}{\text{Number of teeth}}$ [7]

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Q.8 (a) What is nipping in a leaf spring? Discuss its role. List the materials commonly used for the manufacture of the leaf springs. [7]

(b) A helical spring is made from a wire of 6 mm diameter and has outside diameter of 75 mm. If the permissible shear stress is 350 MPa and modulus of rigidity 84 kN/mm^2, find the axial load which the spring can carry and the deflection per active turn. [7]

Q.9 (a) A spur gear has a module of 3 mm, number of teeth 16, a face width of 36 mm and a pressure angle of 20°. It is transmitting a power of 3 kW at 20 rev/s. Taking a velocity factor of 1.5, and Modified Lewis form factor of 0.3, find the stress in the gear tooth. [7]

(b) Write short notes on ***any two*** of the following: [7]

(i) Stress concentration

(ii) Fatigue loading

(iii) Welded joints

(iv) Sliding contact bearing.

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